

Neo-Dualism: Accumulation, Distress, and Proliferation of a Fissured Informality

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Abstract

A dominant strand of orthodoxy argues that the problem of the informal sector could be mitigated through the capitalistic growth process. But our observations on India are different—with an expansion of the capitalistic formal sector, as the economy grows, there is a proliferation of fissured informality. Using a structuralist macro-model, we provide certain explanations for this phenomenon, which are also tested empirically using Indian subnational-state and firm-level data. Thus, we explore both the short- and long-run effects of the expansion of the formal sector on the heterogeneous informal economy. While a section of the population is pulled into the advanced informal activities, a vast segment is pushed to petty production. Accordingly, the orthodox transition narrative is questioned and alternative policy and political possibilities are introduced.

JEL Classification: O11, O13, O17, P48

Keywords

capital accumulation, income deflation, resource expropriation, push and pull effects, employment diversification, migration, informal sector, dualism, nontransition, formal-informal conflicts and complementarities, development macroeconomics, Kalecki, India

1. Introduction

A dominant discourse conjectures that, with the growth of the modern capitalistic formal sector, the informal sector would either wither away or be incorporated into the formal sector or, at least, be able to enter into a phase of tortuous transition (Hymer and Resnick 1969; Ranis and Stewart 1993; Fields 2004;¹ Bardhan 2009; La Porta and Shleifer 2014). However, if we look at figure 1,

¹“[W]e have seen that modern sector enlargement has two beneficial effects on poverty. One is that poverty falls among those workers *who are able to move from the informal sector to the formal sector*. The other is that poverty falls among the workers *who remain informal*. . . . [W]ages do rise in. . . [Lewis]. . . model via these two mechanisms, and poverty is correspondingly reduced *as Lewis-type growth takes place*” (Fields 2004: 13, emphasis added).

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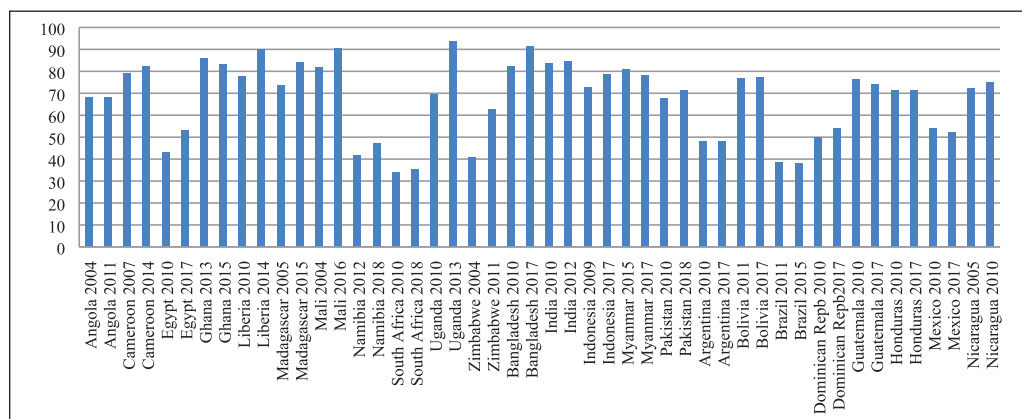


Figure 1. Proportion of informal employment in nonfarm employment (%).

Source: ILOSTAT, International Labor Organization. Retrieved on May 7, 2019, from https://www.ilo.org/ilostat/faces/oracle/webcenter/portallapp/pagehierarchy/Page3.jspx?MBI_ID=540.

which presents the size of nonfarm informal employment of some major countries of Asia, Africa, and Latin America, it becomes clear that the intensity of informal employment is high, and in many countries, it is still increasing. Most intriguingly, gross domestic products of the majority of these countries, especially of Africa and Asia, have grown considerably during or before the periods under consideration (IMF 2019). Further, even among these, **India stands out as a glaring case with very high absolute and relative size of informality, and which is growing, and simultaneously, the growth rate of gross domestic product too is substantially high.** This peculiar feature of the Indian economy grabs our attention. We contemplate whether a study of this Indian scenario could generate some novel as well as generalizable insights on proliferation of informality. Thus, we contemplate to go beyond some recent standard positions which propose informality as a product of lack of sufficient growth and expansion of the capitalist circuit, sufficient education and skill, well-defined property right structure, flexibility of labor market and of governance, sufficient safeguard against global competition, and crisis pushing firms toward informalization, etc. (Portes, Castells, and Benton 1989: 27–29; De Soto 2002; Bardhan 2009; La Porta and Shleifer 2014).

This demands a closer look at the structure of Indian employment and its trends over time, especially when the economy has grown very fast. Table 1, presenting shares of sectoral workforce in the total workforce of India (considering nineteen major subnational states)² over a decade, shows that all the segments of the informal sector (henceforth, INFS:³ with segments such as small-firm/self-employment/own account enterprise⁴ and large-firm/labor-hiring/establishment⁵ across rural-urban locations) are expanding.⁶ But, the share of the agricultural workforce is reducing. Maybe, a part of the workforce moves from agriculture to nonagriculture, mainly to different segments of INFS (over and above crowding of INFS due to population growth expanding the surplus labor force).⁷

²The selected nineteen states are Andhra Pradesh, Assam, Bihar (including Jharkhand), Goa, Gujarat, Haryana, Himachal Pradesh, Jammu & Kashmir, Karnataka, Kerala, Madhya Pradesh (including Chhattisgarh), Maharashtra, Orissa, Punjab, Rajasthan, Tamil Nadu, Tripura, Uttar Pradesh (including Uttarakhand), and West Bengal.

³Here, INFS includes manufacturing and services without construction (as per availability of data).

⁴Own account enterprise without any hired laborer—on a fairly regular basis (NSSO, India).

⁵Enterprise using at least one hired laborer—on a fairly regular basis (NSSO, India).

⁶The authors are grateful to Amit Bhaduri for the idea of tables 1 and 2.

⁷This trend of an increasing volume of INFS continues till 2015–16 for both self-employment and establishments (NSSO, 2015–16).

Table 1. Shares (%) of sectoral workforce in the aggregate workforce in India over time.

Year	Informal rural self-employed firm	Informal rural establishment	Informal urban self-employed firm	Informal urban establishment	Informal sector	Organized manufacturing sector	Agricultural sector	Other sectors
1999–2000	8.40	1.34	5.31	4.03	19.08	2.03	61.01	17.88
2010–11	9.16	2.51	5.99	5.43	23.09	2.71	51.93	22.27

Source: Calculated from different reports of National Sample Survey Office (NSSO), and Annual Survey of Industries—Central Statistics Office (CSO), Government of India.

Note: Other sectors include mainly formal services and formal-informal constructions.

Table 2. Partial labor productivity (annual) in India over time (Rs. at constant 1993–94 price).

Year	Informal rural self-employed firm	Informal rural establishment	Informal urban self-employed firm	Informal urban establishment	Informal sector	Organized manufacturing sector	Agricultural sector	Other sectors
1999–2000	8248	15427	14760	25161	14137	165871	10463	57668
2010–11	10271	21784	17016	30597	18051	299593	14858	100462

Source: Calculated using NSSO, CSO, and Reserve Bank of India database.

Note: Labor productivity is calculated by dividing sectoral gross value added (GVA) with the sectoral workforce.

Other sectors, as above.

Some crucial observations can be derived from table 2 too, which presents partial labor productivity levels of different sectors of India (same nineteen states). The productivity level of formal manufacturing is very high compared to agriculture and INFS, and these absolute differences have been increasing over time. Not only that, the **absolute** differences in productivity levels across INFS segments (rural/urban, self-employment/establishment) have also been increasing.

A **more striking observation** is that the labor productivity level of rural self-employed is lower than all other sectors, even agriculture (table 2), but the share of the workforce in this rural self-employed segment is the largest, and it is increasing (table 1). However, informal establishments, especially the urban ones, are in a better position. So, it seems that, in this process of movement of the workforce away from agriculture, both pull and push factors are working. A part of the agricultural workforce (and of an increased population too) moves into high productivity formal manufacturing and mostly urban INFS establishments, **maybe, due to pull-factors** (like pull from the formal sector, as in Harris and Todaro 1970). On the other hand, perhaps, push-factors (like overcrowding in agriculture, fall in relative income of agriculture, absolute income deflation and forced dissociation of small farmers from means of farming, etc., as in Sanyal 2007) force the agricultural laborers and petty farmers (as well as fresh entrants into the labor force) to get engaged in the less productive rural self-employed segment (and, possibly, in the urban self-employed segment too, as considering the cost of rural-urban migration, effectively, its net per capita value-added could fall below the agricultural average).

Thus, the probable operations of the push- and pull-forces can reallocate the labor, and thereby proliferate the INFS along with its deepening (intra- and intersectoral) divides, strikingly, in a period of high growth. With this crucial backdrop, we posit our basic questions: Is it just a coincidence for India that growth and informality are moving hand in hand? Or, contrary to the orthodox optimism about growth annihilating or absorbing or transforming informality, is it actually a product of growth-pull and associated institutional features (as in Harris and Todaro 1970; Fields 1975;

Thakur and Guha 2019)? Or, is it an outcome of a variety of push factors broadly classified as growth-accumulation-dispossession-informalization nexus (Sanyal 2007; Munck 2013; Standing 2011; Bhaduri 2018)? Or something more nuanced, producing a complex economic structure?

To deal with these issues, first, we introduce a theoretical macro-model along structuralist lines (Kalecki 1934, 1954; Bhaduri 1986; Chakrabarti 2011). The static framework and its short- and long-run variants present succinctly a process of accumulation engendering income-deflation, resource-expropriation as also demand-pull, proliferating a heterogeneous informality, and thereby reproducing a new form of dualism. The specific contributions of this theoretical framework are the following:

- a. It discusses the structural specificities and relationships involving heterogeneous informal and agricultural sectors along with the modern capitalistic formal sector (henceforth, FS).
- b. Contrary to a dominant literature (Ranis and Stewart 1993; Marjit 2003; Fields 2004; Bardhan 2009) presenting a variety of mechanisms through which the FS triggers a transition within the heterogeneous INFS and also brings it close to the FS itself, our model explicates how the FS could, in fact, create informality and thereby, produces not only a new formal-informal dualism (beyond industry-agriculture divide) but also a stark intra-informal duality. Thus, our framework could be a critique of the narrative of transition from traditional to modern.
- c. Going beyond the dichotomous doctrines of growth-pulled (Harris and Todaro 1970; Thakur and Guha 2019) and growth-pushed (Sanyal 2007; Bhaduri 2018) migrations away from the traditional farm, and nonfarm, and thereby creations or expansions of rural-urban INFS, we present a more comprehensive framework combining both the dynamics—pull as also push.
- d. While a large part of the orthodox, as well as heterodox literature, considers informality as an integral segment of the broad capitalistic network, of course having widely diverse loci, relationships, and impacts (Moser 1978; Portes, Castells, and Benton 1989: 26; Basole and Basu 2011; Breman 2013; Munck 2013; Thakur and Guha 2019), an important strand of heterodox literature considers it as a vast sea of surplus humanity that is redundant for the capitalist economic system (Nun 2000; Davis 2004; Sanyal 2007; Chatterjee 2008). We, on the contrary, deal with a dichotomous INFS: the two subsegments with two distinctly different loci vis-à-vis capital; while a relatively advanced but smaller segment is close to capital (and serving it), the remaining vast petty segment is separated (and redundant).
- e. Both the subsegments of our INFS behave in a noncapitalistic way (be it self-employment or hiring labor), survival and decent living being the primary objective of production (and hence, deeply associated with basic resources). Thus, our formal-informal and intra-informal dualities are not just quantitative differences (in terms of size of enterprises). For some of such micro-foundations, we have consulted with a very recent empirical literature (Bhattacharya, Bhattacharya, and Sanyal 2013; Basole, Basu, and Bhattacharya 2015; Chakrabarti 2016; Bhattacharya and Kesar 2018) along with some of the analytical and theoretical writings mentioned earlier.
- f. To the best of our knowledge, no other macroeconomic framework simultaneously discusses a process of creation of informality, its heterogeneity at the micro-level, its heterogeneous loci vis-à-vis capital, and its political-economy implications. Ours is an attempt in this regard.

Subsequently, we test empirically the propositions derived from this model. The fundamental empirical results are:

- ✓ i. There is a clear indication of movement and diversification of workforce from the surplus labor pool and agriculture into both the advanced and petty segments of INFS, respectively.
- ✓ ii. FS has a significant impact on this dual movement/diversification process, most probably, triggering both the push and pull effects.
- ✓ iii. FS productivity improvement increases the productivity gap across advanced and petty segments of the INFS.

Finally, while concluding the paper, certain general political-economic implications of this study are discussed, which could be encapsulated as:

- ✓ i. Modernity itself is creating its own other (outside), reversing the so-called process of transition toward comprehensive capitalism.
- ✓ ii. This creation of a new form of dualism is a particularity of a general global phenomenon: accumulation by dispossession (Harvey 2003) without proletarianization; this makes political-economic “management” of the outside an imperative.
- ✓ iii. Although a part of informality is utilized by (global) capital, a much larger part struggles to procreate itself in limbo. This calls for a dual political-economic response. While the former part of INFS could directly bargain with capital, the latter has to organize (cluster) itself for its fast evaporating spaces—economic-geographic-social-cultural.

2. Structure of Our Model Economy

2.1. Basic features of the sectors

We have a short-run closed economy framework. There are three broad sectors in our abstract economy:

(1) modern capitalistic FS driven by accumulation motive; (2) noncapitalistic INFS with two segments—(a) petty INFS (INFS₁) with small, self-employed firms (mostly rural), devoid of surplus over subsistence and (b) advanced INFS (INFS₂) with large, labor-hiring units (overwhelmingly urban), producing surplus for future consumption over subsistence; (3) as a proxy for the generic resources, a dichotomized agricultural sector—(a) capitalist or modern agricultural sector (MAGR) producing high-value-crops like high-quality cereals, fruits, flowers, vegetables, agro-fuel, feedstock and (b) noncapitalist or traditional agriculture (TAGR) producing mainly coarse food-crops and devoid of surplus. This MAGR-TAGR dichotomy essentially represents modern-traditional duality in (constrained) natural resource use.

Now, FS is characterized by excess-capacity and unemployment (Kalecki 1934). FS produces mainly consumption and investment goods for its workers and capitalists. It also produces inputs (like pesticides, fertilizers, tractors, etc.) and consumer goods (like fast-moving-consumer-goods, garments, consumer durables, etc.) for the agricultural sector as a whole.

While capitalistic MAGR behaves just like FS, noncapitalistic TAGR resembles INFS, as discussed below. Further, due to resource, technology, and institutional constraints, TAGR-output is rigid in the short-run (Kalecki 1954),⁸ although not in the long-run. Essentially, TAGR is a

⁸There is a substantial fall in some of the crucial food crops' growth rates after the liberalization of the Indian economy (table A1, appendix section A3); cereal production across Indian states also falls or increases only marginally over the years (table A2, appendix section A3). The loss of basic food output due to resource diversion might have been countered with an increase through technology etc.; still, per capita food availability has fallen (“In 1991, the per capita food grain availability per annum was 186.2 kg and 177.7 kg in 2016” accessed at: <https://www.google.com/amp/s/www.downtoearth.org.in/news/food/amp/india-claims-to-be-self-sufficient-in-food-production-but-facts-say-otherwise-62091>). Further, aggregate food output might have increased somewhat, but the zones of resource diversion have suffered a loss of critical micro food security (see Chakrabarti and Kundu 2009 and Patnaik 2014: 12–13, for a detailed discussion of the issue; Patnaik apprehending even a macro insecurity).

proxy for the generic resource-constraint that is binding even in the medium-run. Finally, price-determination for TAGR is a complex process that is discussed later.

In INFS, production takes place using indigenously acquired or produced nonfarm consumer goods, inputs, and tools and using surplus labor (especially in the short-run) as also TAGR output (Chakrabarti 2016). INFS output is used by INFS itself and by TAGR.

Advanced INFS₂ product price is cost-determined, however, including a flexible markup on that cost. As demand expands, price increases, raising the markup; this, in turn, induces an expansion of output and employment, given the supply-side conditions, as detailed later.

Petty INFS₁ price is fully cost-determined without any markup, and output is demand-determined, given the supply-side traits (discussed later).

2.2. Linkages among the sectors

The FS and MAGR are assumed to constitute a composite capitalistic entity. In our following analysis, FS actually stands for this composite entity. Conversely, the noncapitalistic INFS is closely associated with noncapitalistic TAGR.

INFS acquires food and other agro raw materials from TAGR and not from MAGR;⁹ conversely, TAGR purchases only INFS₁ products and not the INFS₂ output (of course, a simplifying assumption).

TAGR output is sold to both INFS₂ and INFS₁ through dual institutional arrangements: while TAGR-INFS₂ interaction is dominated by the traders of INFS₂, TAGR-INFS₁ interaction is closed through the open market. Further, a fixed amount of TAGR output is acquired by the traders of INFS₂ at a (pre)contracted fixed price, and it is held as a buffer stock; this stock is released with a fixed premium,¹⁰ as its demand increases. Thus, the actual petty-farmer of TAGR does not have direct access to the INFS₂ market.¹¹ However, TAGR-INFS₁ interactions are much closer: TAGR sells the other part of its output in the open market, which is directly purchased by the INFS₁ for consumption as well as production, at a market-determined flexible price; conversely,

⁹MAGR produces high-value-crop mainly for urban high-end consumers and export. Its inputs and production-harvesting-sorting-cleaning-storage-transportation-processing-packaging-selling processes have to be highly sophisticated and the latter organized through complex supply chain management using modern technology (via a variety of firm-farm contracts led by corporate capital). Consequently, MAGR is integrated with the FS. Contrarily, TAGR is deeply associated with the INFS; the vast INFS population cannot afford to depend on MAGR for production and consumption. For details on this aspect, we could refer to World Bank (2007); Chakrabarti and Kundu (2009); Patnaik (2014).

¹⁰Even if this premium rises with a rise in demand for TAGR output from INFS₂ itself, it should not fundamentally affect our basic structure. In an extreme situation, only the markup may fall in INFS₂, if its price rises, due to increased demand from FS, less than proportionately (compared to the rise in TAGR price).

¹¹“The Dalwai Committee on Doubling Farmers’ Income (Government of India) has pointed out that the share of farmers in consumer’s price is very low; it generally varies from 15 to 40 percent. Studies conducted by the International Food Policy Research Institute and World Bank have confirmed this. The dominant role of middlemen, among others, is primarily responsible for farmers not realizing a reasonable price for their produce, lowering farm income and profitability. This was recognized by the 12th Plan’s Working Group on Agriculture Marketing (India, 2011)” (<https://www.google.com/amp/s/www.thehindu-businessline.com/opinion/agriculture-market-reforms-are-a-must/article10007561.ece/amp/>, downloaded July 10, 2019). Further, the distortionary market power of the rural commercial interests, controlling and depressing the farm-gate price, in particular, is elaborated by Harriss-White (2013) and Rakshit (2014). NSSO, Government of India Report, Situation Assessment Survey of Agricultural Households 2012–13, shows that a large section of the farmers, especially for paddy, sell their produce at a price lower than the government-stipulated minimum support price; and this happens, despite the absence of information asymmetry.

TAGR acquires inputs and outputs directly from the INFS_1 , again at a flexible price. The intersectoral exchange rate and clearing of both the markets are elaborated later.¹²

Next, TAGR has to depend on the FS for modern agricultural inputs (machinery, seeds, fertilizer, pesticides). These inputs are sold to TAGR by the FS via powerful dealers/traders.¹³ TAGR also purchases a variety of consumer goods from the FS (garments, processed foods, cosmetics, and many other fast-moving-consumer-goods items)¹⁴.

On the other hand, we assume away direct interactions between INFS_2 and INFS_1 and between FS and INFS_1 to simplify our framework.

Next, there is a specific type of interaction between the FS and INFS_2 . In reality, INFS_2 uses some FS commodities, but it is assumed away, as most of the commodities used by INFS_2 are indigenously produced (and also due to the fact that the reverse flow of commodities from FS to INFS_2 and hence, the issue of trade balance may create unnecessary complications of macroeconomic accounting). Therefore, we propose that our FS purchases finished products and various types of inputs from the INFS_2 via market exchange as well as subcontracting, primarily to reduce costs and avoid labor resistance. Thus, a constant fraction of FS income is spent on INFS_2 .¹⁵

2.3. A Framework of Formal-Informal-Agriculture interactions

Based on the above-mentioned structures and linkages, we now try to build a comprehensive macroeconomic framework of Formal-Informal-Agriculture interactions, which can be expressed via a flowchart¹⁶ (diagram 1). The corresponding equation system is presented in subsections A1.1, A1.2, A1.3, and A1.4 of appendix section A1.

2.3.1. Expansion of the FS—short-run push and pull effects. Let us first discuss the short-run impacts of an expansion of FS on the sectoral volumes of different segments of INFS . If FS-investment (in nominal terms) rises, the FS expands, raising the level of demand for INFS_2 output. Consequently, INFS_2 product-price increases. It has the following effects:

With a rise in INFS_2 product-price, there is a cost-push increase in FS product-price too. However, the FS price rise only partially offsets the initial investment thrust within FS, as its output is elastic.

On the other hand, as INFS_2 product-price rises, we have: (a) increased utilization of indigenous (surplus) resources by the INFS_2 , (b) increased demand for TAGR output inducing TAGR-supply within INFS_2 through a reduction of the buffer-stock, and (c) increased utilization of (mostly urban) surplus labor. All these processes, arising out of FS-pull, drive up the levels of employment and output in the INFS_2 . Furthermore, although INFS_2 product-price rises, resource and labor prices

¹²This dual market for agriculture with the dual (fix-flex) price is found in the classic paper of Bose (1989).

¹³In the post-green-revolutionary society, it is obvious that both the traditional and modern farmers have to depend on seeds, fertilizer, pesticides, and also equipment produced by the FS. In fact, in many cases, the FS products are thrust upon the TAGR petty-farmers using different types of socio-economic networks and political and technological instruments. A dreadful situation pertaining to chemical fertilizer use could be found in <https://www.downtoearth.org.in/news/agriculture/42-of-india-s-districts-use-85-of-its-chemical-fertilisers-55267> (downloaded July 8, 2020).

¹⁴Its enormous importance could be understood from <https://www.investindia.gov.in/team-india-blogs/fmcg-industry-overview> (downloaded July 24, 2020).

¹⁵The FS acquires cheap resources, wage-goods, and even labor from the INFS_2 to maintain/increase its profit (Bremar 2013; also can be found in Harvey 2003, chap. 4, in a much broader context).

¹⁶There are different types of interactions among the sectors, but here we consider only the dominant streams.

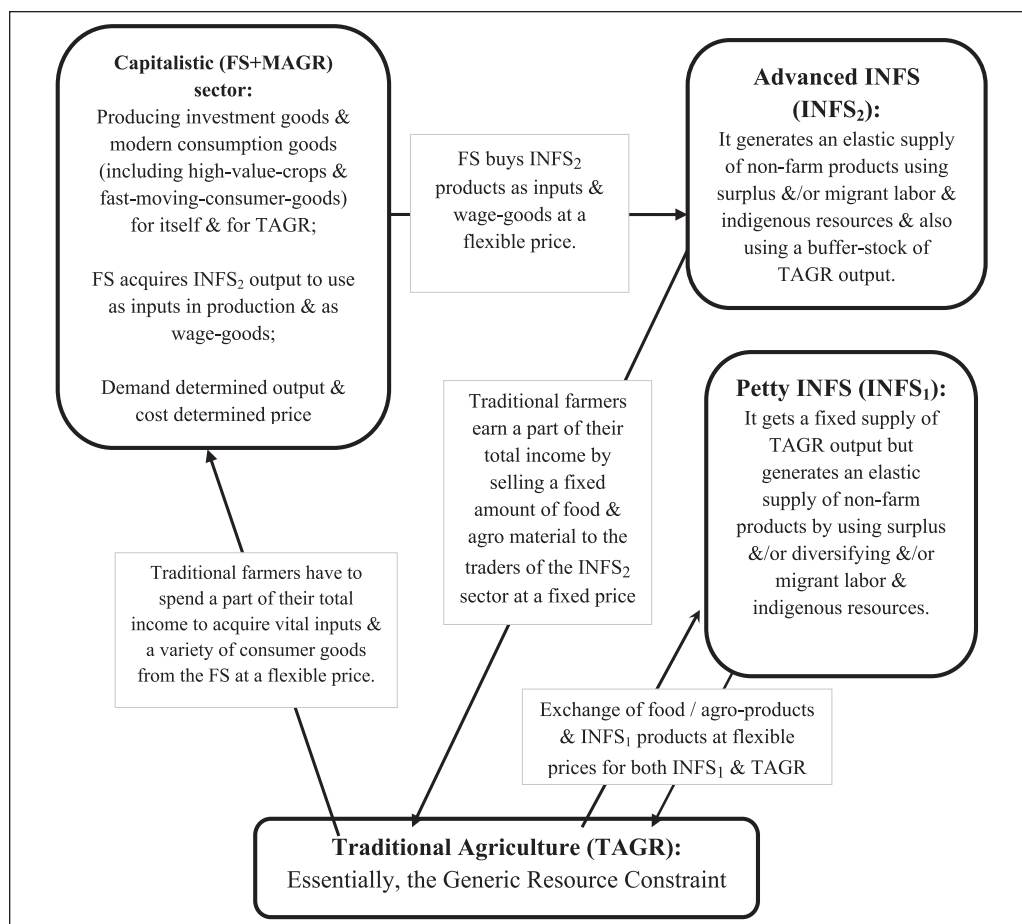


Diagram I. A Flowchart on FS-INFS-agriculture income/expenditure linkages.

remain the same in the presence of surpluses; and TAGR supply-price is structurally set by the traders, as discussed earlier. Consequently, the **INFS₂ gains, in terms of real income too.**

Let us now turn to the INFS₁. **FS price rise** (as stated above) **reduces TAGR-farmers' real income in terms of FS output (that is used for production and consumption in TAGR).**¹⁷ Hence, there is a **localized distressed-diversification from the TAGR to (mostly rural) INFS₁; this localized pushed-diversification of work is undertaken without spatial migration, either replacing farming or supplementing it. This expands the volume of labor supply to the INFS₁, which may encourage its production by using other underutilized indigenous resources. Next, as the FS price rises, a higher portion of the TAGR-farmers' nominal income is spent on the INFS₁ product due to a substitution effect. Hence, the net demand for INFS₁ output also increases** (pulling labor

¹⁷Fixity of the price received by TAGR from INFS₂ intermediaries, as mentioned earlier, is essentially a simplifying assumption. What is important is that, although FS price rises and TAGR production becomes costlier, TAGR farm-gate price cannot increase commensurately due to intermediary presence, despite a demand-pull. Consequently, the terms of trade moves against the petty farmers, leading to income deflation. Such an adverse terms of trade has actually persisted in India even at a time of high growth (see table A3 of appendix section A3). In fact, this kind of adverse terms of trade against agriculture could be beneficial, not only for the traders/intermediaries of the INFS₂ but also for the FS, if there be a tacit understanding between the two powerful lobbies. In that case, the FS too could have indirect access to crucial resources and could, in fact, encourage the commercial interests.

from TAGR as also from the surplus pool). All these push and pull factors drive up the levels of employment and output in the INFS₁.

Next, we bring in the crucial aspect of TAGR supply-constraint for the INFS₁. As the FS expands, INFS₁ employment and output expand as well (as seen just above). But, we know, TAGR-supply to the INFS₁ is given (after deducting the TAGR buffer-stock for INFS₂ from the aggregate TAGR output that is fixed in the short-run). Consequently, if INFS₁ employment has to rise via work-diversification and surplus labor engagement, the per capita intake of TAGR output by INFS₁ has to fall to maintain food-market equilibrium for the INFS₁.

Stated otherwise, when there is a distress-driven expansion of the INFS₁, within the INFS₁ too, there is a deteriorating standard of living, given the TAGR supply-constraint in particular and the overall resource-constraint, in general.

Thus, in our framework, the push and pull processes triggered by the FS proliferates the INFS. Not only that, while its advanced (mostly urban) segment prospers, the petty (majorly rural) section immiserates.

However, these outcomes are conditional upon certain restrictions, which are discussed briefly in the appendix (subsection A1.6).

2.3.2. Some tentative long-run implications. Extending this short-run framework, we can analyze some long-run issues as well. A major question that could be discussed extending the present framework is the widely debated problem of accumulation by dispossession (Harvey 2003; also Sanyal 2007; Dell'Angelo et al. 2017; Bhaduri 2018) and, more importantly, its probable effects on the INFS.

A continuous process of FS expansion requires a large amount of resources (water-forest-land). Hence, resources need to be transferred/reorganized/reconstituted away from traditional uses (like basic agriculture, forestry, age-old nonfarm activity, etc.). Whatever be the process of resource-drain (force-driven expropriation or market-driven conversion), the outcome is obviously a resource-squeeze for the traditional activities. Consequently, the indigenous population, engaged in traditional farm, and nonfarm activities, is forced to migrate toward the newly spreading INFS in and around cities and also in villages (e.g., sizeable expansion of petty trade, transportation, etc.). However, the final destination of this migrant mass largely depends on certain crucial traits of the migrants and the pattern and extent of pull from the expanding FS as well (affecting the different segments of INFS differently). This migration decision of an individual can be expressed by conceptually extending the Harris-Todaro framework. The basic structure of this extended framework and its certain tentative outcomes are briefly discussed in the appendix (section A2).

However, at present, we go back to our short-run framework and the corresponding comparative static outcomes for an empirical analysis.

3. Some Empirical Verification of the Short-Run Analysis: The Case of India

We can put forward three general propositions from the theoretical analysis pertaining to the short-run:

1. When the formality expands, the relatively advanced as well as backward segments of INFS expand through the operations of push and pull factors.
2. These push and pull factors diversify employment and/or drive away labor from TAGR toward INFS alongside absorbing the indigenous surplus labor.
3. Even if the FS progresses, the petty segments of INFS fail to gain in terms of income, although the conditions of the relatively advanced INFS-firms improve to a certain extent.

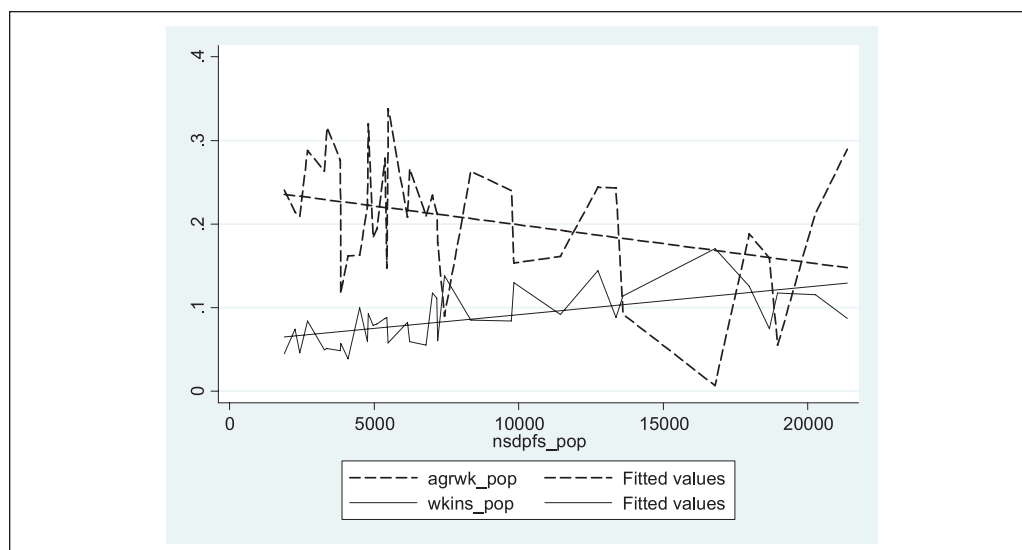


Figure 2. Shares of agriculture and INFS workers (without construction) in total population over per capita net state domestic product of FS²⁰ across 20 major states²¹ of India pooled over 1999–2000 and 2010–11.

Table 3. Corresponding correlation among FS, agriculture, and INFS.

	nsdpfs_pop
agrwk_pop	−0.3444***
wkins_pop	0.5899***

Source: Authors' derivation using NSSO, CSO, and Census data of Government of India.

Note: agrwk_pop and wkins_pop denote shares of agricultural workers and INFS workers in the total population.

nsdpfs_pop denotes per capita net state domestic product of FS.

***denotes 1% level of significance.

First, we present an interesting observation in support of our second premise: an expansion of the economy based on FS growth pushes the distressed agrarian population to diversify and migrate to INFS. The following figures¹⁸ 2 and 3 (and also tables 3 and 4), on India, are self-explanatory. Figure 2 shows, with the growth of the FS, the share of workforce engaged in agriculture is reducing, and the slack is picked up by the INFS. Further, figure 3 shows, with the growth of the FS, the shares of workers in the rural-urban population are expanding for all the segments of INFS.¹⁹

¹⁸Extreme outliers are omitted.

¹⁹The outcomes of figure 3 and table 4 represent the direct impacts (pull) of the FS only. However, push-impact may modify the outcome slightly. A combination of pull and push impact could be captured to some extent through the following regressions.

²⁰NSDP of FS does not include NSDP of construction. NSDP of FS is calculated by adding the GVA of organized manufacturing sector and NSDP formal services. We calculated NSDP formal services by subtracting GVA of informal services from NSDP services, data for which are collected from NSSO reports and Reserve Bank of India website.

²¹Delhi with the other nineteen selected states mentioned earlier.

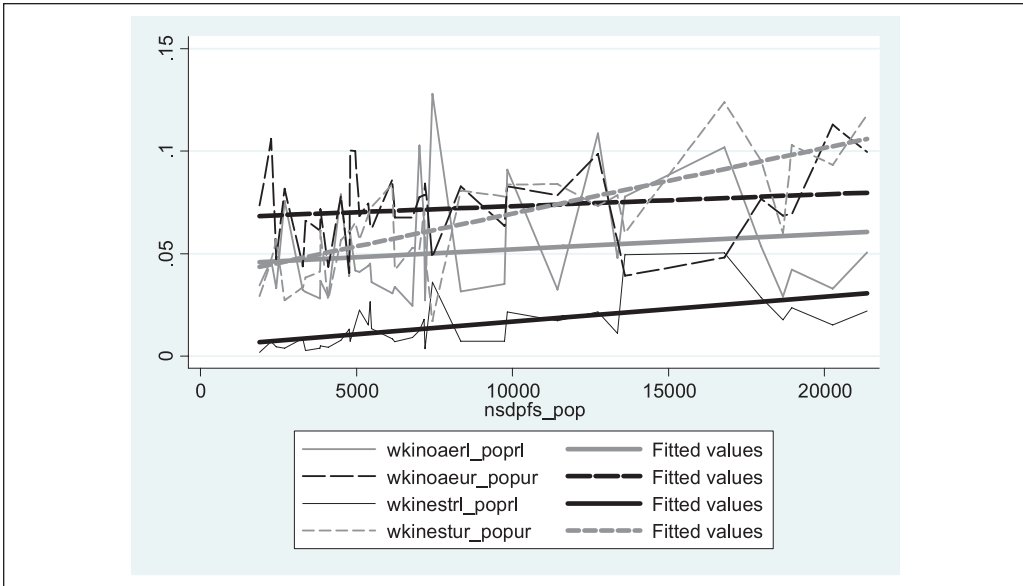


Figure 3. Shares of different segments of INFS worker (without construction) in rural-urban populations over per capita net state domestic product of FS across 20 major states of India pooled over 1999–2000 and 2010–11.

Table 4. Corresponding correlation between FS and different segments of INFS.

	wkinoael_poprl	wkinoaeur_popur	wkinestrl_poprl	wkinestur_popur
nsdpfs_pop	0.1604	0.1734	0.5775***	0.7361***

Source: Same as table 3.

Note: wkinoael_poprl, wkinoaeur_popur, wkinestrl_poprl, and wkinestur_popur denote shares of workers of self-employed rural INFS in rural population, self-employed urban INFS in urban population, rural informal establishment in rural population, and urban informal establishment in urban population, respectively. nsdpfs_pop denotes per capita net state domestic product of FS.

***denotes 1% level of significance.

3.1. Influence of FS expansion on sectoral size of INFS: a state-level analysis

To test proposition 1, we introduce some multiple regression²² models (in table 5) using least-square-dummy-variable (LSDV) method based on secondary data. We consider Indian subnational state level data on unorganized manufacturing sector as a proxy for INFS and formal or organized manufacturing sector data as a proxy for FS.²³ All the required data are collected from

²²For the regressions (tables 5 and 6), necessary pre- and post-estimation tests (such as acprplot with LOWESS for linearity, variance inflation factor for multicollinearity, Histogram test for normality, and insignificance of slope dummies in case of least square dummy variable) have been done. For these, we have used Gujarati and Sangeetha (2004) and the STATA help manual.

²³Most of the Indian literature uses the terms “informal sector” and “unorganized sector” synonymously while using unorganized manufacturing sector data as a proxy for INFS. Following the Indian definitions, there is very little difference between these two terms “informal sector” and “unorganized sector.” Unorganized sector includes all unincorporated proprietary and partnership enterprises (i.e., informal sector enterprises) and enterprises run by cooperative societies, trusts, private and public limited companies (non-ASI). Besides, there are five quinquennial rounds of NSSO data (1984–85, 1989–90, 1994–95, 2000–01, 2005–06) for the unorganized manufacturing sector, while only three rounds of NSSO data for INFS, as a whole. There is also having scarcity of Indian formal service sector data.

Table 5. Regressions²⁶ on sectoral size of rural and urban INFS₁, INFS_{2a}, and INFS_{2b}.

Regression no.	1	2	3	4	5	6
Regressands	Rural INFS ₁ workers	Rural INFS _{2a} workers	Rural INFS _{2b} workers	Urban INFS ₁ workers	Urban INFS _{2a} workers	Urban INFS _{2b} workers
Regressors						
Persons engaged in FS	2.9163*** (.8606)	0.4491*** (.0901)	0.6280*** (.1706)	0.7828*** (.2152)	0.2630*** (.0319)	0.4200*** (.0635)
Share of marginal landholding in total holdings	3952.02 (6531.587)	3797.318*** (613.9244)	1412.592* (721.9344)	1800.629 (1334.919)	-867.7207 (746.5225)	-3002.371** (1309.92)
Share of small landholding in total holdings	54773.02*** (14440.74)	839.541 (1144.121)	4612.865* (2437.59)	5070.628** (2260.923)	785.4429 (1073.39)	3575.603 (2306.666)
Agricultural land	-11.6867 (11.0066)	-0.8871 (1.6679)	-0.3680 (3.2715)	3.2241 (2.9768)	-6.2940*** (2.1456)	-7.3386* (4.1375)
Population	.0162*** (.0029)	.0007 (.0004)	.0011 (.0011)	.0025*** (.0009)	.0020*** (.0005)	.0010* (.0005)
Squared of persons engaged in FS	-2.32e-06*** (6.17e-07)	-2.44e-07*** (6.37e-08)	-3.42e-07*** (1.31e-07)	-3.51e-07*** (1.52e-07)		
Year dummy 1989-90	-23988.77 (186439.1)	-30179.91 (20027.92)		-60081.22 (54015.37)	-18626.13 (21996.51)	
Year dummy 1994-95	-190869.3 (165329.6)	-67299.63*** (21471.13)		-94222.46* (51485.77)	-22597.26 (23155.51)	
Year dummy 2000-2001	-268123.8 (162378.7)	-65799.54*** (20987.16)	22924.29 (26716.68)	-40029.54 (48795.6)	9539.056 (27645.26)	40255.35 (38183)
Year dummy 2005-6	-392451.6** (155477.6)	-56772.67** (23346.55)	14877.56 (30245.9)	-80443.97* (48437.5)	-23786.52 (22435.39)	34174.62 (45225.59)
Constant	-1255737*** (326335.8)	-90947.47*** (27704.31)	-214597.5*** (58592.7)	-180672.6*** (59135.01)	21861.11 (27242.82)	-23199.98 (79692.17)
R ²	0.7265	0.7656	0.6889	0.7844	0.7648	0.6543
Observations	98	99	60	99	99	60

Note: Robust standard errors are in the parentheses.

*, **, and *** indicate 10%, 5%, and 1% levels of significance, respectively.

different sources of Government of India.²⁴ However, we have selected 20 major Indian states, as before, for our analysis.²⁵

In our model, as we proposed to have separate segments of INFS, we consider data on disaggregated INFS as—i) small-sized/self-employment/own-account enterprise; ii) medium-sized/nondirectory-establishment (using 1-5 workers with at least one hired worker on a fairly regular basis); and iii) large/directory-establishment²⁷ (using 6–20 workers with at least one hired worker on a fairly regular basis). We assume, rural-urban medium-sized and large unorganized enterprises²⁸ as advanced/modern INFS or $INFS_2$ and rural-urban small-sized/self-employed enterprises as petty/backward INFS or $INFS_1$ (Basole, Basu, and Bhattacharya 2015; Raj and Sen 2016). Thus, we can view small-sized enterprises as $INFS_1$ and the medium-sized and large enterprises as two different parts of $INFS_2$ viz. $INFS_{2a}$ and $INFS_{2b}$, respectively. However, we need to mention that, with these regression analyses using pooled data, we actually try to understand the intersectoral relations in a better way than simple correlations; with limited secondary data on INFS, we are not claiming any causal relations between the variables.

With these data and methods, we now address the first theoretical proposition, which essentially says, if the FS expands, given the (short-run) total agricultural output, agrarian land-distribution (and hence, the related aspects like input usage, cropping pattern, etc.), and population, all the segments of INFS expand, depending on the relative strengths of pull by the FS and push from agriculture.

The dependent variable of our regression models is the level of employment in different segments of INFS, and the principal regressor is employment in the FS, which is supposed to influence the sectoral size of INFS positively. The control variables are—(a) agricultural land and (b) shares of area of marginal- and small-holdings in the total farming area and (c) population.²⁹

Agricultural land is used as a proxy for agricultural output. It is an important factor having the potential for negatively influencing the sectoral size of INFS; given the relative price of agriculture vis-à-vis FS, as agricultural output falls, farmers are pushed to diversify toward INFS. We cannot control agricultural income (NSDP-agriculture) as a proxy for output because our theoretical model proposes that an expansion of the FS leads to a reduction in farmers' income.

The shares of marginal- and small-landholdings in total farming area are used as a proxy for agrarian land-distribution and also for the extent of traditional crop-farming, which are supposed to positively influence the sectoral size of INFS (as elaborated in Chakrabarti and Kundu 2009:

²⁴The data on the formal/organized manufacturing sector are collected from Annual Survey of Industries, CSO, Govt. of India; the agricultural sector from indiastat.com and Ministry of Agriculture, Govt. of India; population data from Population Census of India; and data on other aggregate activities from Reserve Bank of India database.

²⁵These twenty states cover 95 percent of the unorganized manufacturing sector's employment and GVA in 2005–06 (NSSO 2005–06).

²⁶If we do not include FS into our regression models, the adjusted R^2 decreases by 20 percent to around 50 percent (except in the case of rural $INFS_1$), i.e., in general, the explainability of the models reduces by a large extent. This implies FS is crucial in explaining the behavior of INFS.

²⁷The first two rounds of NSSO data on the unorganized manufacturing sector exclude large enterprises and published data only on small and medium-sized enterprises. The unorganized manufacturing sector data of different sources are not comparable, so the available data on large unorganized manufacturing enterprises from the other sources (like Planning Commission, Government of India) for the pre-liberalization/pre-1991 period are not considered in this analysis.

²⁸They together are called establishments.

²⁹The number of observations, mean, and standard deviation of all the variables are given in table A4 of appendix section A3.

71; Chakrabarti 2016). Lastly, the population of the states is a proxy for state size and also an important positive factor determining the size of INFS workforce.

With these, we can express our regression model as:

$$E_{ist} = \gamma_{1i} + \gamma_{2i}E_{-f_{st}} + \gamma_{3i}Ln_{-ag_{st}} + \gamma_{4i}Sh_{-mg_{st}} + \gamma_{5i}Sh_{-sm_{st}} + \gamma_{6i}P_{st} + \gamma_{7i}TD_1 + \gamma_{8i}TD_2 + \gamma_{9i}TD_3 + \gamma_{10i}TD_4 + u_{ist} \quad (A)$$

i , s , and t stand for i -th segment, s -th state, and t -th year, respectively.

E_{ist} = Employment of the i -th segment (details given below) of INFS in state ' s ' and in year ' t '

$i=1$, if rural INFS₁

$=2$, if rural INFS_{2a}

$=3$, if rural INFS_{2b}

$=4$, if urban INFS₁

$=5$, if urban INFS_{2a}

$=6$, if urban INFS_{2b}

E_{-f} = employment in FS; Ln_{-ag} = total agricultural area or land; Sh_{-mg} = Share of marginal-landholding in total farming area; Sh_{-sm} = Share of small-landholding in total farming area; P = population; TD represents four time-dummies respectively for 1989–90, 1994–95, 2000–2001 and 2005–6. γ_{1i} is the intercept term, and $\gamma_{2i}, \dots, \gamma_{10i}$ are the respective coefficients of the i -th INFS segment; u_{ist} represents the error term.

But equation (A) does not show a linear relationship for all the INFS segments. It shows to have a relationship of quadratic nature between INFS employment and FS employment for the first four INFS segments (rural INFS₁, rural INFS_{2a}, rural INFS_{2b}, and urban INFS₁). It is confirmed by LOWESS (locally weighted scatterplot smoothing). Hence, for these four segments, in the regression equation, a squared of FS employment term is added as:

$$E_{ist} = \gamma_{1i} + \gamma_{2i}E_{-f_{st}} + \gamma_{3i}Ln_{-ag_{st}} + \gamma_{4i}Sh_{-mg_{st}} + \gamma_{5i}Sh_{-sm_{st}} + \gamma_{6i}P_{st} + \gamma_{7i}TD_1 + \gamma_{8i}TD_2 + \gamma_{9i}TD_3 + \gamma_{10i}TD_4 + \gamma_{11i}E_{-f_{st}}^2 + u_{ist} \quad (B)$$

Where $E_{-f_{st}}^2$ = the squared term of the FS employment in state ' s ' and in time ' t '; and ' i ' represents the first four INFS segments only. However, equation (A) holds true for the last two INFS segments, i.e., urban INFS_{2a} and INFS_{2b}.

Table 5 represents our regression results that are mostly in line with our expectations. The regression results show that the state-level sectoral size of INFS across its various subsegments is positively explained by the expansion of FS, controlling for the extent and pattern of farming (land and land distribution) and state population. For the disadvantaged INFS₁, in general, especially rural, we have substantially higher corresponding coefficients than all the other INFS segments. Thus, these regression results (along with figures 2 and 3 of section 3 and tables 1 and 2 of section 1) support the first two propositions in the context of our short-run analysis.

3.2. Influence of FS productivity on labor productivity of INFS: A firm-level analysis

Our next concern is to see whether the progress of FS is able to improve the condition of INFS firms across all its segments or not; i.e., in this subsection, we deal with our third theoretical proposition. For this, we develop some OLS regression models (regressions 1–6 of table 6) based on the latest available firm-level data (NSSO, 2005–6) on the unorganized manufacturing sector

(a proxy for INFS, as before).³⁰ However, we use state-level data for the corresponding sectoral and aggregate economic activities (same data source, as in previous analysis). Once again, we concentrate on 20 major states of India.³¹

As in these regressions, the unit of analysis is firm, and we consider some state-level variables as the key regressors, there is a high chance of having correlation in the error terms within a given state. So, we form clusters for the twenty states and run the regressions with cluster standard error to take care of this within-state correlation that can lead to downwardly biased standard errors.

This study helps to understand the variation in firms' condition (captured through partial labor productivity or GVA per worker) across different segments of INFS with changes in FS activities (i.e., FS labor productivity) when agricultural and state economic activities are controlled.

Thus, the dependent variable³² for these regressions is labor productivity of firms of different segments of INFS and the main regressor is labor productivity of FS. The controlling variables are per capita NSDP (proxy for subnational state's economic condition), share of agricultural NSDP in aggregate NSDP (proxy for agricultural supply), number of workers engaged in INFS firms (proxy for firm-size), and finally, capital/labor ratio of INFS firms (factor intensity). Therefore, we can introduce our regression model as:

$$p_{jis} = \alpha_{1i} + \alpha_{2i} \frac{k_{jis}}{l_{jis}} + \alpha_{3i} l_{jis} + \alpha_{4i} p_f_s + \alpha_{5i} \frac{Y_ag_s}{Y_s} + \alpha_{6i} \frac{Y_s}{P_s} + \varepsilon_{jis} \quad (C)$$

j, i, and s stand for j-th firm, i-th segment, and s-th state, respectively.

p_{jis} = labor productivity of the j-th firm of the i-th segment of INFS in state s.

i = 1, if rural INFS₁

2, if rural INFS_{2a}

3, if rural INFS_{2b}

4, if urban INFS₁

5, if urban INFS_{2a}

6, if urban INFS_{2b}

k = value of the asset of the respective INFS firm; l = number of workers of the respective INFS firm, p_f = state-level average labor productivity of FS firms; Y_ag = NSDP agriculture, Y = NSDP, P = state population; α_{1i} is the intercept term and, $\alpha_{2i}, \dots, \alpha_{6i}$ are the respective coefficients of the variables of the i-th INFS segment; ε_{jis} represents the error term.

First, the general economic condition of a state should have a positive impact on the productivity of INFS. Similarly, a higher share of agricultural NSDP in NSDP implies that the supply of food and agro-raw materials to INFS is relatively abundant, which can enhance INFS labor productivity. A higher share of agricultural NSDP in NSDP also implies a relatively higher income of the farmers and hence, a demand-pull for the INFS products, which can indirectly induce INFS productivity. The number of workers (the most important factor for INFS firms, not only for production but also for organization) and capital/labor ratio are the internal factors of an INFS firm that should positively impact its productivity.

³⁰For this regression analysis, we exclude outliers from the firm-level unorganized manufacturing sector data by using the following method: outliers < (1st Quartile – 1.5*IQR) and outlier > (3rd Quartile + 1.5*IQR); where IQR implies interquartile range.

³¹Here, Bihar does not include Jharkhand, Madhya Pradesh does not include Chhattisgarh, and Uttar Pradesh does not include Uttarakhand. However, we have checked that, even if we incorporate these three new small states into our analysis, the results are almost similar.

³²The details of all the variables are given in tables A5 and A6 of appendix section A3.

Table 6. Regressions on firm-level labor productivity (GVA per worker) across rural-urban INFS₁, INFS_{2a}, and INFS_{2b}.

Regression no.	1	2	3	4	5	6
Regressands	Labor productivity rural INFS ₁	Labor productivity rural INFS _{2a}	Labor productivity rural INFS _{2b}	Labor productivity urban INFS ₁	Labor productivity urban INFS _{2a}	Labor productivity urban INFS _{2b}
Regressors						
Fixed asset per worker of INFS firms	.2098*** (.0159)	.1217*** (.0106)	.1345*** (.0145)	.1630*** (.0085)	.0892*** (.0048)	.1637*** (.0112)
Workers in INFS firms	−663.8405** (259.0514)	597.5508*** (177.5792)	34.8092 (74.6584)	−18.6471 (256.8356)	433.6721*** (94.1299)	1542.784*** (157.2001)
Labor productivity of FS	−.0018 (.0021)	−.0032 (.0023)	.0182*** (.0050)	.0028 (.0019)	.0037** (.0015)	.0078 (.0116)
NSDP agri / NSDP	10017.73* (5467.857)	−2437.721 (7326.037)	−5232.446 (10564.31)	6109.409 (4421.071)	3572.624 (4524.274)	12758.52 (14981.9)
NSDP/ population	.0213 (.0269)	.0435 (.0427)	−.0596 (.0879)	−.0013 (.0194)	.06676*** (.0209)	.1532 (.1153)
Constant	3491.521** (1633.057)	10763.73*** (2120.23)	15446.44*** (4931.077)	3274.286** (1354.154)	8762.507*** (1459.572)	−790.1838 (5438.025)
R ²	0.1232	0.0935	0.1390	0.1719	0.1262	0.2075
Observations	23626	5262	3001	18695	9136	4127

Note: Cluster standard errors are in the parentheses.

*, **, and *** indicate 10%, 5%, and 1% levels of significance, respectively.

Now, regressions of table 6 show that the productivity of FS has a significant positive association only with rural INFS_{2b} and urban INFS_{2a}. However, it does not have any significant positive relation with the other INFS segments; rather, it may have some negative relation (though not significant) with rural petty INFS. Following our theoretical proposition 3, thus, we can say that the rural and urban petty INFS segments are unable to gain from an improvement in labor productivity of FS, while the INFS large-establishments may get some benefit from it. Therefore, FS activity itself may create a duality within the INFS.

Further, when for all the other INFS segments, larger firm-size shows a positive relation with labor productivity, the self-employed INFS shows a negative relation. Stated otherwise, an increase in the number of workers in self-employed enterprises implies congestion within such firms, lowering their productivity level. This happens, maybe, because, distressed farmers throng the petty INFS.

Thus, regression tables 5 and 6 (along with the observations of tables 1 and 2 and of figures 2 and 3) support the outcomes of our theoretical model. They together show an intriguing phenomenon: an expansion of the FS not only engenders a spread of the INFS but may also create a divide within it.

4. In Lieu of Conclusion: Some General Political-Economic Implications

This paper essentially questions the orthodox narrative of the comprehensive capitalistic transition of the Global South. From the perspective of informality, it tries to show how, instead of engendering the so-called inclusive growth and eventual structural transformation, capital itself, in its pursuit of profit, in/directly and involuntarily/deliberately creates its devalued other/outside; only a small part of which is utilized as well as induced by capital, while the other much larger part proliferates in limbo gasping for spaces. Thus, capital itself produces a neo-dualism—a fractured informality—beyond the historical/traditional one, through income-deflation, resource-expropriation as well as devalued-accommodation alongside exclusion. These FS-INFS and intra-INFS dualities are also aggravated by the policies of a neoliberal State, having distinct bias toward big corporations, adversely affecting the interests of TAGR farmers and of petty INFS population.

These heterogeneous phenomena are not only a particularity for a country like India but also could be observed on a global scale blurring the national boundaries, especially in the developing countries like, Indonesia (Rothenberg et al. 2016), Madagascar (Nordman, Rakotomanana, and Roubaud 2016), Ivory Coast (Günther and Launov 2012), Chile (Amuedo-Dorantes 2004), etc.

This complexity of the emerging local/global economic structure challenges the orthodox policy institutions which try to address/redress this neo-dualism mostly with the instruments of development management (essentially, disciplining the outsiders with a variety of doles). Contrarily, we suggest a dual strategy from the perspective of INFS:

While the formal-informal linkages/negotiations/bargaining could be beneficial for the advanced segments of the informal sector (taking advantage of the dependence of FS on the INFS), the petty segments have to fight with the annihilating power of capital, and in that struggle, cluster form of organization could offer a challenge/respite.

However, for both these strategies, political organization, particularly of/for the ‘informal’ should play a crucial unifying role, in conjunction with the general struggle of the working class. This political organization/movement should take up a dual role: on the one hand, pushing the State to introduce policies toward building formal-informal linkages and also formal clusters of informal firms; on the other, orchestrate union/political bargaining with the FS and politically organize the scattered micro-firms for clustering.

Appendix

Section A1: Working of the Interlinked Sectors of Our Short-Run Closed-Economy Model

Subsection A1.1. The basic notations

1. FS (including MAGR) output: Y_i
2. FS employment: L_i
3. FS money-wage: w_m
4. Price of FS output: p_i
5. Fixed nominal investment in FS: I^0
6. INFS₂ output: Y_2
7. Price of INFS₂ product: p_2
8. Surplus per unit of INFS₂ output: s
9. Price of INFS₁ product: p_1
10. Aggregate external demand from the FS for the *net* output of INFS₂: D_2
11. Supply of INFS₂ net-output: S_2
12. Aggregate external demand from the TAGR for the *net* output of INFS₁: D_1
13. Supply of INFS₁ net-output: S_1
14. Fixed/pre-set amount of TAGR-output sold to the big traders of INFS₂: f^0
15. Fixed contracted price of f^0 : p_f^0
16. Fixed TAGR-output: F^0
17. Market determined price of the TAGR output sold to INFS₁: p_{f1}
18. Fraction of TAGR farmers' income spent on the FS output: r
19. Constant amount of labor needed to produce one unit of FS output: a_{li}^0
20. Constant amount of INFS₂ input needed to produce one unit of FS output: a_{2i}^0
21. Constant (aggregate) amount of INFS₂ output needed per unit of FS output: b_{2i}^0
22. Constant amount of INFS₂ output needed (as inputs and consumption goods for labor) to produce a unit of INFS₂ output: b_{22}^0
23. Constant amount of INFS₁ output needed (as inputs and consumption goods for labor) to produce a unit of INFS₁ product: b_{11}^0
24. Fixed amount of TAGR-product used as food (for labor) and nonfood input (for production) per unit of INFS₂ output: a_{f2}^0
25. Amount of TAGR-product used as food (for labor) and nonfood input (for production) per unit of INFS₁ output: a_{f1}

Subsection A1.2. Output and price determination in FS

With excess capacity and unemployment in the short-run, the formal sector's output/real-income (Y_i) is demand-determined and hence:

$$Y_i = \left\{ (w_m / p_i) \cdot L_i \right\} + \left(I^0 / p_i \right) - \left\{ (p_2 \cdot b_{2i}^0 \cdot Y_i) / p_i \right\} + \left[r (p_i / p_1) \cdot \left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} / p_i \right] \quad (1)$$

Workers and capitalists of the FS purchase inputs and outputs from both FS and INFS₂. So, we have the total real spending of the workers and capitalists on the FS output as their aggregate real spending $\left[\left\{ (w_m / p_i) \cdot L_i \right\} + \left(I^0 / p_i \right) \right]$ minus their total real spending on the INFS₂ i.e., $\left\{ (p_2 \cdot b_{2i}^0 \cdot Y_i) / p_i \right\}$.

A fixed/preset amount of TAGR-output (f^0) is sold to the traders of the INFS₂ at a fixed/pre-contracted price (p_f^0); this is used by the INFS₂ itself for consumption and production. The other part of the short-run TAGR-output, i.e., ($F^0 - f^0$) is sold to the INFS₁ at a market-determined price p_{f1} ; $p_{f1} < p_f^0$. Thus, aggregate income of TAGR is $\left[p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right]$.

A part (r) of this TAGR income is spent on the FS for modern inputs and FMCG having a price p_i . Another part $(1 - r)$ of this TAGR income is spent on the INFS₁ for petty consumables at a price p_1 . Now, $r = r(p_i / p_1)$; with, $r^1 < 0$ (subsequently, superscript with '1' indicates the first-order derivative); as (p_i / p_1) rises, r falls. Hence, $\left[r(p_i / p_1) \cdot \left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} / p_i \right]$ represents the amount of FS output demanded by TAGR.

Further, under excess capacity and unemployment in the FS, output/labor ratio, Y_i / L_i is constant and equal to 1 (by suitable choice of unit). Hence, equation 1 can be rewritten as:

$$Y_i = \left\{ (w_m / p_i) \cdot Y_i \right\} + \left(I^0 / p_i \right) - \left\{ (p_2 / p_i) \cdot b_{2i}^0 \cdot Y_i \right\} + \left[r(p_i / p_1) \cdot \left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} / p_i \right] \quad (2)$$

Next, assuming markup pricing by the FS-firm, the FS-price is: $p_i = t \cdot [w_m \cdot a_{li}^0 + p_2 \cdot a_{2i}^0]$

Here, fixed markup $t > 1$ and t captures the average fixed cost, the average cost of non-labor input derived from the FS itself, and average profit. We also assume FS money-wage is indexed to INFS₂ product-price.

Hence, $w_m / p_2 = e$ (a constant).

Again, we assume, for simplicity, $a_{li}^0 = a_{2i}^0 = 1$

$$\text{Hence, } p_i = t \cdot [e \cdot p_2 + p_2] = t \cdot (1 + e) \cdot p_2 \quad (3)$$

So, $p_2 / p_i = 1 / \{t \cdot (1 + e)\} = d$ (a constant).

This implies $p_i = p_2 / d$; and also

$$w_m / p_i = w_m / (p_2 / d) = d \cdot e \text{ (as } w_m / p_2 = e, \text{ as above)} = c \text{ (a constant)}$$

Hence, replacing (w_m / p_i) with c , (p_2 / p_i) with d and p_i with (p_2 / d) in equation 2, we have FS-output:

$$Y_i = c \cdot Y_i + d \cdot I^0 / p_2 - \left\{ d \cdot b_{2i}^0 \cdot Y_i \right\} + \left[r \left\{ p_2 / (d \cdot p_1) \right\} \cdot \left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} \cdot d / p_2 \right]; \quad \text{with } r^1 < 0.$$

Finally,

$$Y_i^* = \left[\left(d \cdot I^0 / p_2 \right) + r \left\{ p_2 / (d \cdot p_1) \right\} \cdot \left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} \cdot d / p_2 \right] / (1 - c + d \cdot b_{2i}^0) \quad (4)$$

Hence, ceteris paribus, $Y_i^* = Y_i(p_2)$; with, $Y_i^1 < 0$

As p_2 falls, p_i falls as well (given $p_2 / p_i = d$, from equation 3); this induces—(a) real investment in the FS and (b) the TAGR farmers' real income as well as its share that is spent on the FS. All these boost the demand (and so, supply) for FS output.

Subsection A1.3. Price and output determination in INFS₂

First, INFS₂ can generate a surplus over and above the cost of production. Alongside own product, TAGR output is also essential for both worker and entrepreneur of INFS₂.³³

Given these traits, INFS₂ price is: $p_2 = s + (p_2 \cdot b_{22}^0) + (p_f^0 \cdot a_{f2}^0)$; $s \geq 0$

From the above price-equation, we get, $s = s(p_2 / p_f^0)$; with, $s^1 > 0$.

Now, we take up the issue of output determination in this INFS₂.

Aggregate external demand from the FS for the net output of INFS₂ (i.e., net of intrasectoral use) is:

$$D_2 = b_{2i}^0 \cdot Y_i = b_{2i}^0 \cdot \left[\left[(d \cdot I^0 / p_2) + r \{ p_2 / (d \cdot p_1) \} \cdot \{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \} \cdot d / p_2 \right] / (1 - c + d \cdot b_{2i}^0) \right]$$

$$= D_2(p_2); \text{ with, } D_2^1 < 0 \quad (5)$$

This implies an inverse relation between D_2 and p_2 (as in diagram A1). As p_2 falls, FS expands via a fall in p_i (as mentioned above). These expansionary boosts for FS increase the demand for INFS₂ too. Hence, we have a downward sloping D_2 curve. On the other hand, with a rise in FS-investment, ceteris paribus, FS expands, and demand for INFS₂ (D_2) rises as well, shifting the D_2 curve upward.

Now, D_2 determines the net output of INFS₂, which gives rise to the corresponding gross output as well (via b_{22}^0), given the following supply-side considerations:

There is no limiting capital in the INFS₂; nonfarm inputs, including minor implements and nonfarm consumer goods, are produced indigenously; and there is an availability of surplus labor. As a result, the INFS₂ output becomes demand-determined, provided there is a sufficient supply of TAGR products. Given these conditions, along with the assumption of the buffer stock, a rise in p_2 would have the following repercussions:

- Indigenous (surplus) resources of INFS₂ are used more extensively.
- Indigenous (surplus) labor is used more extensively.
- Buffer-stock of TAGR-output held by the traders of INFS₂ is reduced, and TAGR-output supply rises within the INFS₂. It helps in increasing the employment and output of INFS₂. It happens as the amount of TAGR-product used per unit of INFS₂ output (a_{f2}^0) remains fixed, and the TAGR-supply rises.

Given these, S_2 , the supply of INFS₂ net-output (net of intrasectoral use) responds positively to a rise in p_2 . Hence, $S_2 = S_2(p_2)$; with, $S_2^1 > 0$, as shown in diagram A1.

The interactions between D_2 and S_2 in diagram A1 determine the INFS₂ equilibrium price (p_2^*) and the equilibrium net output (S_2^*) at E_0 . From this equilibrium net output, the corresponding gross output could be derived by using the factor b_{22}^0 , as mentioned earlier.

Subsection A1.4. Price and output determination in INFS₁

The INFS₁ firms are not capable of earning a surplus over and above the cost of production (i.e., cost of own output used for production and self-consumption and cost of TAGR output used as input and consumption-good). And only a residual (fixed) amount ($F^0 - f^0$) of TAGR output is available for them at a market-determined (flexible) price p_{f1} .

³³An overwhelming majority of the Indian INFS firms (especially the manufacturing ones) are found to be agro-based. For details, see Chakrabarti (2016: 182).

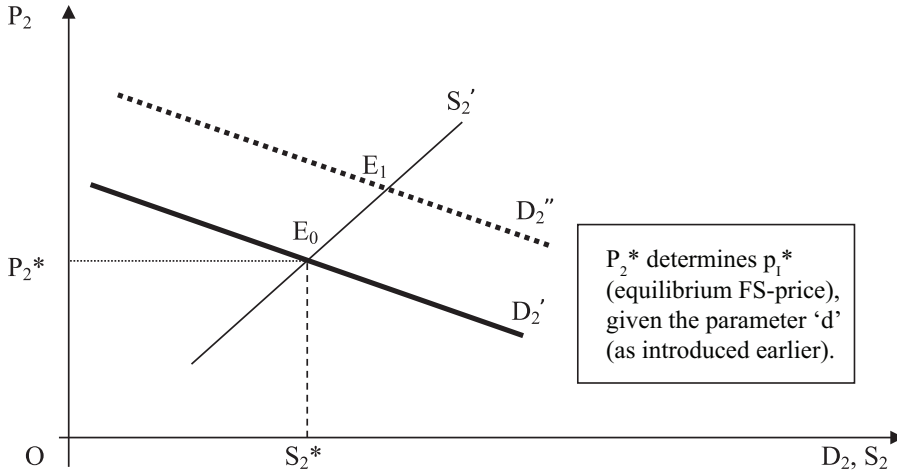


Diagram A1. Market for INFS₂ and comparative static outcomes.

Given these traits, the INFS₁ price (p_1) is:

$$p_1 = (p_1 \cdot b_{11}^0) + \{p_{f1} \cdot a_{f1} (p_1 / p_{f1})\}; \text{ with, } a_{f1}^1 > 0 \{a_{f1} \text{ rises with a rise in } (p_1 / p_{f1})\}.$$

$$\text{Hence, } (p_1 / p_{f1}) = \left[\{a_{f1} (p_1 / p_{f1})\} / (1 - b_{11}^0) \right] \quad (6)$$

The structural difference between INFS₂ and INFS₁ is that, while the INFS₂ is able to maintain $a_{f2} = a_{f2}^0$ and with a fall in p_2 , only the average surplus 's' falls, in the INFS₁, with a fall in p_1 , a_{f1} has to fall.

Now, we take up the issue of output determination in INFS₁. Aggregate external demand from TAGR for net output of INFS₁ (i.e., net of intrasectoral use) is:

$D_1 = [\{1 - r(p_i / p_1)\} \{p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0)\} / p_1]$; with $r = r(p_i / p_1)$ and $r^1 < 0$, as discussed in the context of equation 1.

$$\text{Hence, } D_1 = D_1(p_1); D_1^1 < 0. \quad (7)$$

As p_1 falls, the real income of TAGR-farmer and its share that is spent on INFS₁ both rise; hence, we get a downward sloping D_1 curve. On the other hand, as p_i rises, r falls, and hence, $(1 - r)$ rises; a higher portion of TAGR-farmers' nominal income is spent on INFS₁ product. Consequently, with a rise in p_i , D_1 rises, and the corresponding curve shifts to the right, as in diagram A2.

Next, D_1 determines the net output in INFS₁, which gives rise to the corresponding gross output as well (via b_{11}^0).

Let us propose a function for the net-supply of output in INFS₁ (net of intrasectoral use) as: $S_1 = S_1(p_1)$; with, $S_1^1 > 0$. As p_1 rises, the indigenous resources and labor of INFS₁ are used more extensively, and the impoverished farmers, locally diversifying away from TAGR (due to a reduction in their real income³⁴ in terms of INFS₁-output), are also pulled; consequently,

³⁴As TAGR farmers have to purchase the INFS₁ products for production and consumption purposes.

production in INFS_1 expands and the level of its net-output rises. On the other hand, if FS-price p_i rises, ceteris paribus, TAGR-farmers' real income in terms of FS-output falls.³⁵ This farmers' distress leads to localized employment-diversification from TAGR towards INFS_1 . Consequently, the supply of net-output, i.e., S_1 , increases, and the corresponding curve shifts to the right, as in diagram A2.

The interactions between D_1 and S_1 determine the INFS_1 equilibrium price (p_1^*) and the equilibrium net output (S_1^*), at E_0 . From this equilibrium net output, the corresponding gross output could be derived by using the factor b_{11}^0 , as mentioned before.

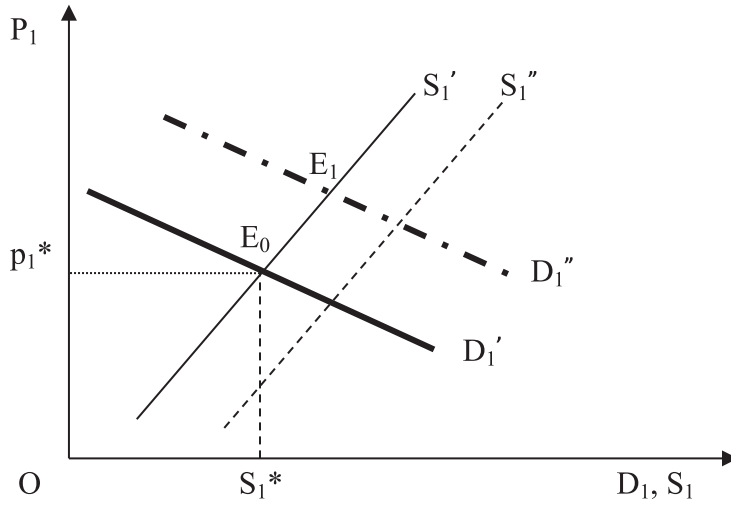


Diagram A2. Market for INFS_1 and comparative static outcomes.

Subsection A1.5. Impact of expansion of FS: some short-run implications

Here, we try to explain in brief, how FS can create the dual effects (as described in section 2.3.1) on the INFS using diagrams A1, A2, and A3. If FS expands, the level of demand for INFS_2 output increases. This shifts the D_2' curve up to D_2'' in diagram A1, leading to an increase in the price of INFS_2 (p_2). This increase in p_2 induces supply of INFS_2 output by using indigenous surplus labor and resources and by drawing of TAGR output from the buffer stock.

In contrast, diagram A2 shows what happens to INFS_1 employment and income if there is a cost-push increase in p_i (due to an increase in p_2 , as in the case just above) leading to a fall in the real income in TAGR. The farmers and laborers diversify their work (locally) towards INFS_1 . Consequently, the net supply of INFS_1 output increases (by using surplus resources along with the diversifying and indigenous surplus labor). That is, the S_1 curve is pushed to the right from S_1' to S_1'' in diagram A2. On the other hand, increase in p_i increases the demand for INFS_1 output from TAGR due to a substitution effect. Hence, the net demand-curve for INFS_1 output, i.e., D_1' curve too shifts to the right. Consequently, output and employment must rise. However, the price of INFS_1 product (p_1) may rise/fall/remain the same depending on the extent of shifts in D_1 and S_1 (as is clear from diagram A2). But, this is the case when we ignore the supply-constraint put by TAGR for INFS_1 .

³⁵As TAGR farmers have to purchase modern inputs and consumer merchandise from the FS.

Hence, we introduce a TAGR supply-constraint as in the short-run, only $(F^0 - f^0)$ amount of TAGR output is available for INFS_1 . So, if INFS_1 -employment (L_1) has to rise, per unit requirement of TAGR output by INFS_1 (i.e., a_{f1}) has to fall to maintain food-market equilibrium for the INFS_1 ; hence, the terms-of-trade between INFS_1 and TAGR (i.e., p_1 / p_{f1}) has to fall. This is clear from the food-market equilibrium condition for INFS_1 :

$$(F^0 - f^0) = \{a_{f1}(p_1 / p_{f1}) \cdot L_1\}; \text{ with } a_{f1}^1 > 0 \quad (8)$$

Thus, a rise in p_1 has to be less than the rise in TAGR-price (p_{f1}); p_{f1} rises due to an increased demand for TAGR-output, given the expansion of output and employment in the INFS_1 and given the TAGR-supply for INFS_1 . This could be captured by diagram A3 through a shift in the food market equilibrium from E_0 to E_1 .

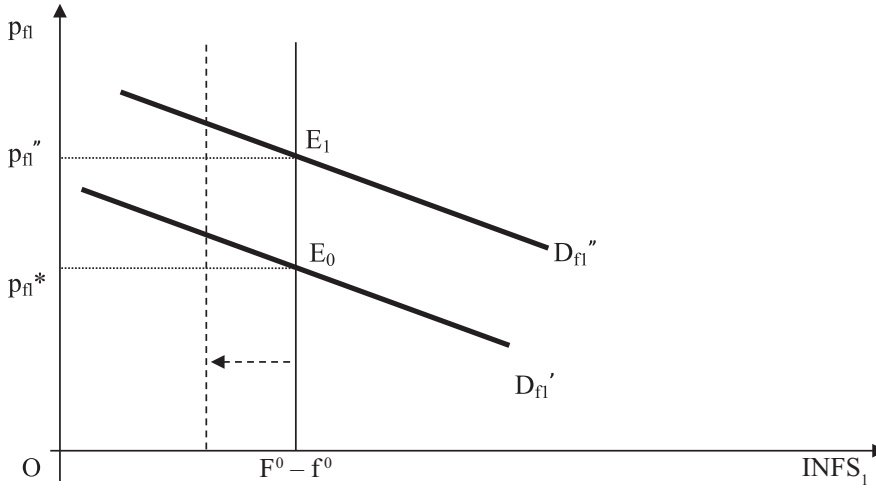


Diagram A3. Food-market equilibrium for the INFS_1 .

Subsection A1.6. Some heuristic calculations

Finally, summarizing the above discussion, we have three markets (INFS_1 , INFS_2 , and TAGR markets) with the following three equilibrium conditions:

a. From equation 5 and the subsequent discussion:

$$S_2 = S_2(p_2) = D_2 = b_{2i}^0 \cdot Y_i = b_{2i}^0 \cdot \left[\left[(d \cdot I^0 / p_2) + r \{ p_2 / (d \cdot p_1) \} \cdot \{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \} \cdot d / p_2 \right] / (1 - c + d \cdot b_{2i}^0) \right] \\ \text{with } r^1 < 0 \quad (9)$$

b. From equation 7 and the corresponding discussion;

$$S_1 = S_1(p_1) = D_1 = \left[\{ 1 - r(p_i / p_1) \} \cdot \{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \} / p_1 \right]; \text{ with } r^1 < 0 \quad (10)$$

c. From equation 8 along with equation 5 and its subsequent discussion:

$$(F^0 - f^0) = [a_{f1}(p_1 / p_{f1}) \cdot L_1(p_i, p_1, p_{f1})]; \text{ with } a_{f1}^1 > 0 \quad (11)$$

And, we have the derived relationship from equation (3):

$$p_i = p_2 / d \quad (12)$$

These four equations (9 through 12) solve for four variables p_2, p_i, p_1, p_{f1} . Consequently, the equilibrium values of outputs and employments of FS, INFS₁, and INFS₂ could be derived and subsequently, the impacts on these outcomes due to changes in the exogenous variables and parameters could be derived.

Let us start with an increase in private investment in the FS.

This would expand FS and hence INFS₂, provided the expansionary impact of this investment on FS (captured by the term $(d \cdot I^0 / p_2)$ of equation 9) is larger than the contractionary effect on FS (captured by the rest of the equation 9) arising from TAGR owing to FS price rise due to an INFS₂ price increase. In fact, this condition seems to be likely from the general experience.

Next, this process would also induce INFS₁ if the value of the right-hand side of equation 10 increases despite a rise in p_1 (as a consequence of an increase in FS investment, as discussed earlier). From equation 11, i.e., equation 8 and its corresponding discussion), p_{f1} / p_1 rises despite a rise in p_1 . Consequently, the value of the term $\left[\left\{ p_{f1} \cdot (F^0 - f^0) + (p_f^0 \cdot f^0) \right\} / p_1 \right]$ of equation 10 rises unambiguously. However, the value of the other term $\{1 - r(p_i / p_1)\}$ of equation 10 may reduce, as, theoretically, p_i / p_1 may fall (although both p_i and p_1 rise with a rise in FS investment). However, this fall in p_i / p_1 and hence that of $\{1 - r(p_i / p_1)\}$ should have a limit if INFS₁ has to expand. This limit is likely, as the rise in p_1 cannot be large, given that INFS₁ can expand, in the face of rising demand from TAGR, by using surplus labor and indigenous resources.

Section A2: Few Long-Run Implications

Subsection A2.1. Notations

Average income of an entrepreneur in urban advanced INFS = $I_2^{u,e}$

Probability of becoming an entrepreneur in urban advanced INFS = $P_2^{u,e}$

Average income of a wage laborer in urban advanced INFS = $I_2^{u,w}$

Probability of becoming a wage laborer in urban advanced INFS = $P_2^{u,w}$

Average income of a self-employed working-owner in urban petty INFS = I_1^u

Probability of becoming a self-employed working-owner in urban petty INFS = P_1^u

Average income of an entrepreneur in rural advanced INFS = $I_2^{r,e}$

Probability of becoming an entrepreneur in rural advanced INFS = $P_2^{r,e}$

Average income of a wage laborer in rural advanced INFS = $I_2^{r,w}$

Probability of becoming a wage laborer in rural advanced INFS = $P_2^{r,w}$

Average income of a self-employed working-owner in rural petty INFS = I_1^r

Probability of becoming a self-employed working-owner in rural petty INFS = P_1^r

Subsection A2.2. The basic framework

The processes of FS expansion, expansion of the FS-linked INFS, extraction of resources from the traditional economy (i.e., our TAGR), distress and even dispossession, migration (of this distressed or dispossessed) from the TAGR to the dichotomous INFS (advanced/establishment

and petty/self-employed INFS), and thereby proliferation of a fissured informality (through push and pull thrusts) could operate through the following channels.

Let us start with a situation where FS expands, extracting resources from the TAGR and thereby, suffocating a section of the TAGR population. This dispossessed mass has the following options in the long-run (assuming they don't have options to get engaged in the FS):

1. If these people do not have the urban network, 'urban skills', and access to credit, they are bound to get engaged in the rural economy either as a wage laborer in the advanced INFS or as a petty self-employed (if they have some access to resources and markets). In both the situations, most probably, their condition deteriorates.
2. If, however, these people do not have the urban network and urban skills but have access to credit, resources, and markets, they could transform into entrepreneurs in the rural advanced INFS.
3. If, on the other hand, these people have the urban network and 'urban-skills' as well as access to credit, resources, and markets, their employment options (in rural-urban INFS) could be expressed by invoking the concept of migration equilibrium as introduced by Harris-Todaro (1970; see Debraj Ray 1998, for a simplified version). We introduce this long-run migration equilibrium condition (involving the rural advanced INFS and the advanced and petty segments of urban INFS) as:

Expected urban income = Expected rural income

$$\text{i.e., } I_2^{u,e} \cdot P_2^{u,e} + I_2^{u,w} \cdot P_2^{u,w} + I_1^u \cdot P_1^u = I_2^{r,e} \cdot P_2^{r,e}$$

These probabilities could be dependent on the existing employment opportunities across the different segments of INFS (and it is true for all the following cases).

4. Next, if these evicted people have the urban network and 'urban-skills', but don't have access to credit, resources, and markets, the migration equilibrium condition changes as:

$$I_2^{u,w} \cdot P_2^{u,w} = I_2^{r,w} \cdot P_2^{r,w}$$

5. If, on the other hand, these people have the urban network and urban-skills as well as limited access to resources and markets (although not to credit), the migration equilibrium condition becomes:

$$I_2^{u,w} \cdot P_2^{u,w} + I_1^u \cdot P_1^u = I_2^{r,w} \cdot P_2^{r,w} + I_1^r \cdot P_1^r$$

6. Now, combining all these options the (rural-urban) migration equilibrium condition can be expressed as

Expected urban income = Expected rural income

$$\text{i.e., } I_2^{u,e} \cdot P_2^{u,e} + I_2^{u,w} \cdot P_2^{u,w} + I_1^u \cdot P_1^u = I_2^{r,e} \cdot P_2^{r,e} + I_2^{r,w} \cdot P_2^{r,w} + I_1^r \cdot P_1^r$$

Here (in all the above cases), these probabilities depend on the patterns and extents of the urban network, urban skills, and access to credit, resources, and markets for the potential migrants as

also on the pattern and extent of the existing employment opportunities across the different segments of INFS.

Given these options and migration equilibrium conditions (3 to 6), as FS expands and (via linkages) drives up the activities of the rural-urban advanced INFS (although not of the petty INFS), the left-hand side and right-hand side of the above equations are affected, and the two may become unequal. The direction of this inequality would then guide the dispossessed population to take a decision on migration (whether to find an appropriate job in the rural economy or to migrate to urban).

7. Finally, we introduce a situation where FS expands, extracting resources from the TAGR and thereby squeezing a section of the TAGR population. This disadvantaged mass (although not dispossessed yet) has the following long-run migration equilibrium condition:

Expected INFS income = Average income in TAGR

$$\text{i.e., } I_2^{u,e} \cdot P_2^{u,e} + I_2^{u,w} \cdot P_2^{u,w} + I_1^u \cdot P_1^u + I_2^{r,e} \cdot P_2^{r,e} + I_2^{r,w} \cdot P_2^{r,w} + I_1^r \cdot P_1^r \\ = \text{Average income in TAGR}$$

Given this condition, as FS expands and (via linkages) drives up the activities of the rural-urban advanced INFS (although not of the petty INFS) and also suffocates the TAGR, the left-hand side rises although the right-hand side falls, triggering migration towards INFS. The destination of the migrants would largely depend on the initial conditions of the migrants (as discussed above) as also on the pattern and extent of pull from the different segments of INFS. However, migration itself would generate congestion across INFS and act as a counterbalancing force in the course of time.

These broad frameworks (grounded on micro decision making but having macro implications) and outcomes (micro as well as macro) could be made more structured through modeling exercise. However, we refrain from that, for the paucity of space. In fact, this could be an important extension of the present study.

Section A3: Some Relevant Tables

Table A1. Economic reforms and the rate of growth of agricultural output.

	Growth rate (%)	
	Preliberalization period (1969–70 to 1992–93)	Postliberalization period (1993–94 to 2016–17)
Rice	2.91	1.31
Wheat	4.60	1.92

Source: Directorate of Economics and Statistics, Ministry of Agriculture, Government of India.

Table A2. Annual average growth rate (%) of cereal production across 15 major states of India between 2002–3 and 2012–13.

States	Small & marginal farmers	Other farmers	Total
Andhra Pradesh	8.47	5.51	7.35
Assam	4.96	-2.98	3.01
Bihar	-0.04	-4.78	-1.28
Gujarat	3.21	-2.33	0.97
Haryana	5.47	-0.61	1.74
Karnataka	0.72	2.20	1.40
Kerala	-7.11	10.77	-3.78
Madhya Pradesh	4.81	-0.19	2.16
Maharashtra	-0.31	-3.94	-1.99
Orissa	5.49	6.64	5.78
Punjab	-2.40	-1.24	-1.48
Rajasthan	7.90	3.05	5.77
Tamil Nadu	-0.56	-1.86	-0.98
Uttar Pradesh	1.80	-0.20	1.35
West Bengal	-3.26	-7.40	-3.93
Total (15 states)	1.71	-0.80	0.77
All India	1.68	-1.28	0.58

Source: Computed from NSSO 59th and 70th rounds unit-level data on the Situation Assessment Survey of Farmers.

Table A3. Index of terms-of-trade between farmers and nonfarmers from 2004–5 to 2015–16 (base: triennium ending 2011–12 = 100).

Years	Index of terms-of-trade between farmers and nonfarmers
2004–5	87.78
2005–6	84.77
2006–7	87.05
2007–8	92.22
2008–9	99.98
2009–10	100.13
2010–11	102.95
2011–12	97.27
2012–13	97.4
2013–14	98.6
2014–15	97.58
2015–16	96.96

Source: Agricultural Statistics at a Glance, Directorate of Economics and Statistics, Ministry of Agriculture, Government of India, 2017.

Table A4. Mean and standard deviation (SD) of the variables used for the state-level analysis.

Variables	Observations	Mean	SD
Workers in rural INFS ₁	98	900794.6	991660.8
Workers in rural INFS _{2a}	99	108751.5	114393.2
Workers in rural INFS _{2b}	60	139253.8	162409.4
Workers in urban INFS ₁	100	268157.3	274319.4
Workers in urban INFS _{2a}	100	158368.1	146114
Workers in urban INFS _{2b}	60	176849.6	211418.4
Employment or persons engaged in FS	100	420635.8	369466.2
Share of area of marginal landholdings in total land holdings (in %)	99	22.82	15.09
Total agricultural land (in '000 hectares)	100	8014.8	7421.62
State population (in lakh)	100	456.03	389.02

Source: Calculated based on different NSSO reports, ASI reports, indiastat.com website, and population census and excluding extreme outliers.

Table A5. Mean and standard deviation of the firm-level variables used.

Variables	Observations	Mean	SD
GVA per worker in rural INFS ₁	26302	6721.69	5158.91
GVA per worker in rural INFS _{2a}	5845	15135.45	8746.91
GVA per worker in rural INFS _{2b}	3217	21821.7	13713.12
GVA per worker in urban INFS ₁	20498	7999.98	6519.13
GVA per worker in urban INFS _{2a}	9939	16146.66	8134.48
GVA per worker in urban INFS _{2b}	4295	30529.86	23087.03
FA per worker in rural INFS ₁	26302	7852.356	7639.03
FA per worker in rural INFS _{2a}	5845	22336.23	20260.26
FA per worker in rural INFS _{2b}	3217	32154.86	32230.42
FA per worker in urban INFS ₁	20498	16086.66	16008.85
FA per worker in urban INFS _{2a}	9939	33902.88	27579.91
FA per worker in urban INFS _{2b}	4295	55519.97	49469.93
Workers in rural INFS ₁	26302	1.56	0.71
Workers in rural INFS _{2a}	5845	3.03	1.08
Workers in rural INFS _{2b}	3217	9.71	3.98
Workers in urban INFS ₁	20498	1.56	0.76
Workers in urban INFS _{2a}	9939	3.21	1.09
Workers in urban INFS _{2b}	4295	9.10	3.39

Source: Calculated based on 2005–6 firm-level data of NSSO, India, and removing outliers.

Table A6. Mean and standard deviation of the state-level variables used for the firm-level analysis.

Variables	Observations	Mean	SD
Average labor productivity in FS	20	226833.2	158907.2
Share of agricultural NSDP in total NSDP	20	0.24	0.09
NSDP per capita	20	14074.46	7520.14

Source: Calculated based on ASI reports, RBI website, and population census.

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